

TissueForge AI

The Operating System for Tissue Engineering & Regenerative Medicine

Executive Summary

TissueForge AI is the autonomous intelligence platform accelerating the \$150B+ regenerative medicine revolution. We're building the AI-native operating system that transforms tissue engineering from artisanal science to predictable manufacturing — enabling lab-grown organs, bioprinted tissues, and personalized regenerative therapies at scale.

Every 10 minutes, someone dies waiting for an organ transplant. Over 100,000 Americans languish on transplant lists while 17 die daily. Meanwhile, billions suffer from chronic conditions that could be solved by regenerative therapies — if only the technology could scale.

TissueForge AI provides the complete intelligence layer: AI-designed tissue scaffolds, real-time bioreactor optimization, quality prediction, regulatory documentation automation, and the industry's first comprehensive tissue development knowledge graph.

Investment: \$25M Series A

Valuation: \$125M

Market: \$150B+ (regenerative medicine, tissue engineering, bioprinting, organ transplantation)

Traction: 48 biotech partners, 12 research hospitals, \$8.4M ARR, 23 tissues in development pipeline

The Problem

The Organ Shortage Crisis

The human cost is staggering: - 106,000+ Americans on organ transplant waiting lists - 17 people die every day waiting for transplants - Only 36,000 transplants performed annually in the US - 8,000+ people removed from lists yearly — too sick to survive - Average wait time: 3-5 years for kidneys, 4+ years for hearts - \$35B annual cost of dialysis alone (while patients wait for kidneys)

Why traditional solutions fail: - Organ donation plateaued — not enough donors - Animal organs (xenotransplantation) face immune rejection - Artificial organs still primitive and life-limiting - The math doesn't work: demand grows 5x faster than supply

The Tissue Engineering Bottleneck

The science works — but scaling doesn't:

Challenge	Current State	Impact
Development time	10-15 years per tissue type	Too slow to save lives
Success rate	5-10% of attempts produce viable tissue	Massive waste
Reproducibility	High variability between batches	Regulatory nightmare
Cost	\$1M+ per engineered organ	Economically impossible
Expertise	<1,000 qualified tissue engineers globally	Severe bottleneck
Quality control	Manual inspection, post-facto testing	Defects discovered too late
Bioreactor optimization	Trial and error over months	Lost time and materials

The core problems:

- 1. Design Complexity** - Tissues contain billions of cells in precise 3D arrangements - Multiple cell types must interact correctly - Vascularization (blood vessel growth) remains unsolved for thick tissues - Each patient's biology requires customization
 - 2. Manufacturing Chaos** - Bioreactors are "black boxes" — limited real-time visibility - Small environmental changes cascade into failures - No predictive models for tissue development trajectories - Batch failures discovered after weeks of cultivation
 - 3. Regulatory Wilderness** - FDA pathways unclear for engineered tissues - Documentation requirements massive and manual - No standardized quality metrics - Each submission essentially novel
 - 4. Knowledge Fragmentation** - Research siloed across institutions - Protocol variations make replication difficult - Failures rarely published — lessons lost - No shared intelligence layer
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The TissueForge Solution

The Complete Intelligence Stack for Regenerative Medicine

TissueForge AI provides an autonomous platform that transforms tissue engineering from artisanal craft to predictable, scalable manufacturing.

Core Platform Components

1. TissueDesigner AI — Generative Scaffold Intelligence AI-powered tissue architecture design:

- **Multi-scale modeling:** Design from molecular to organ level
- **Vascularization optimization:** AI-generated blood vessel networks
- **Cell placement intelligence:** Optimal positioning for each cell type
- **Patient customization:** Designs adapted to individual biology
- **Bioprinting instruction generation:** Direct-to-printer files

Capabilities: - Generate scaffold designs 100x faster than manual methods - Predict vascularization success before printing - Optimize for mechanical properties, porosity, degradation - Integrate patient imaging (CT/MRI) for personalized geometries - Library of 500+ validated scaffold architectures

2. BioreactorMind — Real-Time Cultivation Intelligence Autonomous bioreactor optimization:

- **Sensor fusion:** Integrate 50+ parameters (pH, O₂, CO₂, glucose, lactate, temperature, flow, pressure, impedance, imaging)
- **Predictive modeling:** Forecast tissue development trajectory
- **Anomaly detection:** Identify problems hours/days before failure
- **Autonomous control:** AI-optimized environmental parameters
- **Digital twin:** Complete simulation of tissue development

Impact: - 40% reduction in batch failures - 3x faster development cycles - 60% improvement in tissue viability - Real-time quality prediction (vs. post-facto testing)

3. QualityLens — Intelligent Quality Assurance AI-powered quality control:

- **Non-destructive imaging analysis:** Real-time structural assessment
- **Viability prediction:** Cell health scoring without sampling
- **Mechanical property estimation:** Predict strength, elasticity
- **Functional assessment:** Predict tissue performance
- **Release criteria automation:** Objective quality gates

Capabilities: - Continuous quality monitoring throughout cultivation - Predict final quality from early-stage indicators - Automated documentation of quality parameters - Statistical process control for manufacturing consistency

4. RegulatoryPilot — Compliance & Documentation Intelligence Autonomous regulatory navigation:

- **IND/BLA preparation:** Auto-generate submission documents
- **CMC documentation:** Chemistry, Manufacturing, Controls automation
- **Audit trail:** Complete traceability from design to delivery
- **FDA pathway optimization:** Recommend optimal regulatory strategies
- **Global harmonization:** Support FDA, EMA, PMDA requirements

Impact: - 70% reduction in regulatory preparation time - Automated batch records and documentation - Real-time compliance monitoring - Accelerated approval timelines

5. TissueGraph — The Industry Knowledge Layer The world's most comprehensive tissue engineering intelligence:

- **Protocol database:** 50,000+ validated protocols
- **Failure analysis:** Learnings from 200,000+ experiments
- **Material intelligence:** Biomaterial properties and compatibility
- **Cell behavior models:** Predictive models for 2,000+ cell types
- **Literature integration:** Real-time research synthesis

Technology Deep Dive

The TissueForge AI Architecture

TissueForge AI Platform

TissueDesigner	BioreactorMind	QualityLens	RegPilot
Generative	Real-Time	QA/QC	Compliance
Design	Control	Intelligence	Engine

TissueGraph Knowledge Layer

Protocols • Materials • Cell Models • Research • Failures

Foundation AI Models

Tissue Transformers • Physics-Informed Neural Networks
Multi-Modal Vision • Temporal Sequence Models

Integration Layer

Bioprinters • Bioreactors • Imaging Systems • LIMS • ERP

Key Technical Innovations

1. **Tissue Transformers** Our proprietary foundation models for biological systems:
 - **Training data:** 500M+ cell observations, 100K+ tissue development sequences
 - **Multi-scale representation:** Molecular → cellular → tissue → organ
 - **Physics-informed architecture:** Conservation laws embedded in model structure
 - **Temporal modeling:** Predict development trajectories over weeks/months
2. **Vascularization Intelligence** The breakthrough that enables thick tissue engineering:
 - **Generative vessel networks:** AI-designed vascular trees
 - **Perfusion optimization:** Ensure all cells receive nutrients
 - **Angiogenesis prediction:** Model blood vessel growth dynamics
 - **Patient-specific adaptation:** Integrate with existing vasculature
3. **Real-Time Bioreactor Vision** Advanced sensing and interpretation:
 - **Multi-spectral imaging:** See beyond visible light
 - **Optical coherence tomography integration:** 3D tissue visualization
 - **Cell tracking:** Monitor millions of cells simultaneously
 - **Fusion architecture:** Combine imaging with biochemical sensors
4. **Federated Learning Network** Learn from the entire industry while protecting IP:
 - **Privacy-preserving training:** No raw data leaves facilities
 - **Model aggregation:** Combine learnings across 100+ sites
 - **Differential privacy:** Mathematical guarantees on data protection
 - **Competitive separation:** Learnings shared, specifics protected

Market Opportunity

The Regenerative Medicine Explosion

Total Addressable Market: **\$150B+ by 2030**

Segment	2024	2030	CAGR
Regenerative Medicine	\$25B	\$80B	21%
Tissue Engineering	\$15B	\$45B	20%
Bioprinting	\$2B	\$8B	26%
Cell Therapy	\$10B	\$35B	23%
Organ Transplantation	\$12B	\$20B	9%
Total	\$64B	\$188B	20%

Key Market Drivers

1. **Organ shortage crisis worsening** — demand growing 3x faster than supply
2. **Bioprinting technology maturing** — hardware now capable, software lagging
3. **Regulatory pathways clarifying** — FDA advancing guidance
4. **Investment surge** — \$15B+ invested in cell/gene therapy 2023-2025
5. **Aging populations** — degenerative diseases driving demand
6. **Manufacturing shift** — moving from academic to commercial scale

Customer Segments

Tier 1: Biotech Companies (\$50B market) - Tissue engineering startups - Cell therapy developers - Regenerative medicine companies - Organ bioengineering ventures

Tier 2: Research Hospitals & Academic Medical Centers (\$30B market) - Transplant centers - Regenerative medicine institutes - Clinical trial sites - Translational research labs

Tier 3: Pharmaceutical & Medical Device Companies (\$40B market) - Drug delivery system developers - Implant manufacturers - Wound healing product companies - Bioprinting equipment makers

Tier 4: Contract Development & Manufacturing (\$20B market) - CDMOs entering regenerative medicine - Cell therapy manufacturers - Specialty biologics producers

Business Model

Revenue Streams

1. Platform Subscription (60% of revenue)

Tier	Annual Cost	Included
Research	\$50K	5 users, basic modules, TissueGraph access
Development	\$150K	25 users, full platform, 3 tissue programs
Manufacturing	\$400K	Unlimited users, production modules, priority support
Enterprise	\$1M+	Multi-site, custom integration, dedicated success team

2. Compute & AI Services (25% of revenue)

- **Design generation:** \$5K-50K per tissue design
- **Simulation runs:** \$500-5K per simulation
- **Quality analysis:** \$200 per batch assessment
- **Regulatory document generation:** \$20K-100K per submission package

3. Success Fees & Licensing (15% of revenue)

- **Milestone payments:** \$100K-500K upon tissue development milestones
- **Royalties:** 1-3% on tissues developed using platform
- **IP licensing:** Platform-generated designs available for license

Unit Economics

Metric	Value
Average Contract Value (ACV)	\$280K
Gross Margin	82%
CAC	\$85K
CAC Payback	4 months
Net Revenue Retention	145%
LTV:CAC	12:1

Competitive Landscape

The Competitive Matrix

Capability	TissueForge AI	Traditional LIMS	Academic Tools	Manual Processes
AI-powered design	Generative		Basic	
Real-time bioreactor control	Autonomous			
Quality prediction	Predictive	Reactive		
Regulatory automation	Full stack	Templates		
Knowledge graph	Comprehensive		Siloed	
Federated learning	Industry-wide			

Why We Win

1. **First comprehensive platform** — competitors offer point solutions
2. **AI-native architecture** — built for intelligence, not retrofitted
3. **Data network effects** — each customer improves platform for all
4. **Regulatory expertise** — team includes ex-FDA reviewers
5. **Hardware agnostic** — integrate with any bioprinter/bioreactor

Traction & Validation

Platform Metrics

Metric	Current
Biotech partners	48
Research hospital deployments	12
ARR	\$8.4M
ARR growth (YoY)	280%
Tissues in development pipeline	23
Protocols in TissueGraph	50,000+
Bioreactor hours monitored	2.4M

Customer Highlights

Organovo — Bioprinted liver tissue leader - “TissueForge reduced our development cycles by 60%” - 3 tissue programs on platform - \$1.2M annual contract

United Therapeutics — Organ manufacturing pioneer - Platform deployed across 4 facilities - Lung tissue program accelerated 18 months - Strategic partnership discussions underway

Mayo Clinic — Regenerative medicine powerhouse - Clinical research deployment - 8 research programs using platform - Co-development agreement on cardiac tissue

Aspect Biosystems — Bioprinting innovator - TissueDesigner integrated with printing systems - OEM partnership in discussion - Joint product development

Scientific Validation

- 12 peer-reviewed publications citing TissueForge
 - 4 patents granted, 11 pending
 - NIH SBIR Phase II: \$2.5M
 - DARPA contract: Battlefield tissue manufacturing
 - FDA Breakthrough Device designation: 2 tissues
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Go-to-Market Strategy

Phase 1: Research Adoption (Current)

Target: Academic labs and early-stage biotechs **Strategy:** Free TissueGraph access, research tier pricing

Goal: Build knowledge network, establish credibility

Phase 2: Development Scale (2026-2027)

Target: Clinical-stage biotech companies **Strategy:** Development tier, regulatory modules **Goal:** Demonstrate clinical impact, expand use cases

Phase 3: Manufacturing Domination (2027-2028)

Target: Manufacturing-scale operations, CDMOs **Strategy:** Enterprise deployments, production modules

Goal: Become industry standard platform

Phase 4: Ecosystem Platform (2028+)

Target: Entire regenerative medicine industry **Strategy:** Platform services, marketplace, data products

Goal: Network effects moat

Sales Motion

Research Tier: Product-led growth, self-serve **Development Tier:** Inside sales, 30-day trials **Manufacturing/Enterprise:** Field sales, executive engagement

Team

Leadership

Dr. Sarah Chen — CEO & Co-Founder - PhD Biomedical Engineering, MIT - Former VP R&D, Organovo - 15 years tissue engineering experience - Led development of first FDA-approved bioprinted tissue

Dr. Marcus Williams — CTO & Co-Founder - PhD Computer Science (AI/ML), Stanford - Former Staff Research Scientist, DeepMind - Pioneer in AI for biological systems - 50+ publications, 12 patents

Dr. Jennifer Park — Chief Science Officer - MD/PhD, Johns Hopkins - Former Director, Regenerative Medicine, NIH - Board-certified in surgery - Led landmark clinical trials in tissue transplantation

David Rodriguez — Chief Revenue Officer - 20 years enterprise software sales - Former VP Sales, Benchling (led to \$100M ARR) - Expert in life science software GTM - Built teams across US, EU, Asia

Dr. Michael Chang — Chief Regulatory Officer - PhD Biomedical Engineering - 12 years at FDA, CBER division - Led review of multiple tissue engineering products - Author of FDA tissue engineering guidance

Advisory Board

- **Dr. Robert Langer** — MIT Institute Professor, tissue engineering pioneer
- **Dr. Jennifer Doudna** — Nobel Laureate, CRISPR pioneer
- **Dr. Anthony Atala** — Director, Wake Forest Institute for Regenerative Medicine
- **George Church** — Harvard Professor, synthetic biology leader
- **Vijay Pande** — Andreessen Horowitz General Partner

Team Composition

- 85 employees across 4 locations
- 60% hold advanced degrees (PhD/MD)
- Average 12 years industry experience
- Engineers from: Google, DeepMind, Genentech, Organovo
- Scientists from: MIT, Stanford, Harvard, Johns Hopkins

Financial Projections

Revenue Forecast

Year	ARR	Growth	Customers	Employees
2025	\$8.4M	—	60	85
2026	\$24M	186%	150	140
2027	\$58M	142%	350	220
2028	\$120M	107%	700	350
2029	\$220M	83%	1,200	500

Path to Profitability

- **Gross margin:** 82% (software + managed services)
- **Operating margin target:** 25% at scale
- **Break-even:** Q4 2027 at \$45M ARR
- **Free cash flow positive:** 2028

Use of Funds (\$25M Series A)

Category	Amount	Purpose
R&D	\$12M	Foundation model development, new modules
Sales & Marketing	\$7M	Enterprise sales team, customer success
Operations	\$4M	Infrastructure, regulatory, quality
G&A	\$2M	Legal, finance, HR

Investment Opportunity

The Ask

\$25M Series A at \$125M pre-money valuation

Why Now

1. **Technology inflection point** — AI capabilities finally match tissue complexity
2. **Market ready** — Bioprinting/bioreactor hardware mature, software lagging
3. **Regulatory clarity** — FDA pathways solidifying
4. **Urgent need** — Organ shortage crisis demands solutions
5. **First mover advantage** — No comprehensive platform exists

What We've Proven

- Technology works — 48 production deployments
- Customers pay — \$8.4M ARR, 280% growth
- Science valid — 12 peer-reviewed publications
- Team executes — Former leaders from Organovo, DeepMind, FDA
- Market exists — \$150B+ opportunity

Exit Potential

Strategic Acquirers: - Major pharma (J&J, Roche, Novartis) - Medical device companies (Medtronic, Boston Scientific) - Tech giants entering healthcare (Google, Apple, Microsoft) - Emerging regenerative medicine leaders

Comparable Transactions: - Benchling: \$6.1B valuation (lab software) - Recursion: \$4B valuation (AI for drug discovery) - Insitro: \$2.4B valuation (AI biology)

IPO Path: \$200M+ ARR by 2029 supports public offering

The Vision

2035: The TissueForge Future

Universal organ availability: - No one dies waiting for a transplant - Personalized organs printed in days
- Regenerative therapies for all degenerative diseases

TissueForge as critical infrastructure: - Platform behind 80% of engineered tissues - Global quality standards powered by our AI - \$10B+ market cap, category-defining platform

Healthcare transformed: - Disease treated through regeneration, not management - Personalized medicine extended to tissue level - Lifespan extended through organ replacement

Next Steps

Immediate Priorities (6-12 months)

1. **Close Series A** — \$25M to accelerate growth
2. **Launch manufacturing modules** — Production-grade capabilities
3. **FDA breakthrough path** — First platform-developed tissue in trials
4. **Expand to Europe** — EMA regulatory modules, EU sales
5. **Strategic partnerships** — Major biotech/pharma deals

Contact

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“We’re not just building software — we’re building the infrastructure for humanity’s next chapter in medicine. A future where no one dies waiting for an organ. Where tissue damage is repaired, not managed. Where regeneration replaces degradation. TissueForge AI is the foundation of that future.”

— Dr. Sarah Chen, CEO